

Package ‘DrugSigNet’

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Type Package

Title Drug Signature Network

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Description R package to conduct in silico drug repurposing analysis by leveraging several algorithms for both signature-based and network-based approaches.

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URL <https://github.com/compneurobio/DrugSigNet>

BugReports <https://github.com/compneurobio/DrugSigNet/issues>

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Alluvyplot	<i>Plot Alluvial Diagram (robust version)</i>
------------	---

Description

Creates an alluvial diagram from categorical data with multiple axes. Supports columns with special characters, duplicate rows, optional frequency column, and multiple file types.

Usage

```
Alluvyplot(
  object = NULL,
  data_df,
  axis_columns,
  freq_col = NULL,
  file_type = "pdf",
  file_name,
  width = 20,
  height = 15
)

## S4 method for signature 'PlotObject'
Alluvyplot(object, axis_columns, freq_col = NULL)
```

Arguments

object	Optional PlotObject. If NULL, a new one is created from data_df.
data_df	Data frame containing categorical variables.
axis_columns	Character vector of column names to use as axes (minimum 2).
freq_col	Optional column representing frequency/weight. Defaults to 2 if not provided.
file_type	Output file type: "pdf", "png", "jpeg", etc. (default: "pdf").
file_name	Name of the output file (without extension).
width	Plot width (default: 20 cm).
height	Plot height (default: 15 cm).

Value

A ggplot object representing the alluvial diagram.

B3DB_classification	<i>B3DB blood–brain barrier (BBB) permeability classification dataset</i>
---------------------	---

Description

A curated dataset of experimentally measured blood–brain barrier (BBB) permeability information for small molecules, derived from the **Brain-Blood Barrier Database (B3DB)**.

Usage

```
B3DB_classification
```

Format

A tibble with 7,807 rows and 12 variables:

row_id Row index from the original B3DB dataset

compound_name Common or preferred compound name

IUPAC_name IUPAC chemical name

SMILES SMILES representation of the compound

CID PubChem Compound Identifier (character)

logBB Logarithmic blood–brain barrier partition coefficient

BBB_class Binary BBB permeability classification (BBB+ or BBB-)

Inchi InChI string for the compound

threshold Threshold value used for BBB classification, if available

reference Reference codes corresponding to source publications

group Grouping label defined by the original authors

comments Additional comments or annotations

Details

The dataset compiles chemical structures, identifiers, and quantitative BBB permeability measurements (logBB), along with a binary BBB permeability classification. The data are suitable for cheminformatics, pharmacokinetics, and machine learning applications related to CNS drug discovery.

This dataset was obtained from the supplementary materials of:

```
> Meng et al. (2021), > *A curated diverse molecular database of blood–brain barrier permeability  
> with chemical descriptors*, > **Scientific Data**, 8, 310.
```

BBB classification

The column `BBB_class` provides a categorical annotation indicating whether a compound is considered permeable (BBB+) or non-permeable (BBB-) to the blood–brain barrier, based on experimental logBB values and literature-defined thresholds.

Data preprocessing

The dataset has been minimally processed for compatibility with R and package development best practices:

- Column BBB+/BBB- was renamed to BBB_class.
- Column NO. was renamed to row_id.
- PubChem compound identifiers (CID) are stored as character strings to preserve identifier integrity.

No filtering, relabeling, or modification of the original experimental measurements or classifications has been performed.

Source

Meng, F., Xi, Y., Huang, J., et al. (2021). *A curated diverse molecular database of blood–brain barrier permeability with chemical descriptors*. Scientific Data, 8, 310. <https://www.nature.com/articles/s41597-021-01069-5>

calc_drug_similarity *Calculate structural similarity among drugs*

Description

Computes pairwise structural similarity between drugs using canonical SMILES and molecular fingerprints. Drug names are resolved to canonical SMILES using a selected reference database before similarity calculation.

Usage

```
calc_drug_similarity(  
  object = NULL,  
  drugs,  
  source = c("OpenTargets", "ChEMBL", "B3DB_classification"),  
  fingerprint = "standard",  
  method = "tanimoto"  
)
```

Arguments

object	A DrugAnnotation object. If NULL, a new object is created internally.
drugs	A character vector or a one-column data frame containing drug names.
source	Character string specifying the reference database used to map drug names to canonical SMILES. One of "OpenTargets", "ChEMBL", or "B3DB_classification".
fingerprint	Character string specifying the fingerprint type passed to rdck . Common options include "standard", "extended", and "pubchem".
method	Similarity metric to use. Currently supports "tanimoto".

Details

Structural similarity is computed using fingerprint-based methods (default: CDK standard fingerprints) and the Tanimoto similarity metric.

The resulting similarity matrix is stored as an attribute ("similarity_matrix") of the result data frame.

Similarity values range from 0 (no structural similarity) to 1 (identical structures). InChIKeys are not supported because they are hashed identifiers and do not preserve structural similarity.

At least two drugs with valid canonical SMILES must be available to compute similarity.

Value

A DrugAnnotation object. The result slot contains a data frame with drug names and canonical SMILES. The similarity matrix can be accessed via: `attr(x@result, "similarity_matrix")`.

CHEMBL*ChEMBL drug-level annotation dataset*

Description

A curated, drug-centric dataset derived from the ChEMBL v36 SQLite database. The dataset integrates drug mechanisms of action, molecular targets, UniProt accessions, ATC classifications, chemical structures, clinical indications, and safety warnings.

Usage

CHEMBL

Format

A data frame with one row per ChEMBL drug and the following variables:

drug_chembl_id ChEMBL drug identifier

drug_name Preferred drug name

drug_synonym Drug synonyms (semicolon-separated, order preserved)

syn_type Synonym types

max_phase Maximum clinical development phase of the drug

canonical_smiles Canonical SMILES representation

standard_inchi_key Standard InChIKey

drug_indication Clinical indications (semicolon-separated, ordered)

mesh_id MeSH identifiers aligned with drug_indication

efo_id EFO identifiers aligned with drug_indication

max_phase_for_ind Maximum phase per indication, aligned by position

target_chembl_id ChEMBL target identifiers

target_name Target names

target_type Target types

organism Target organism

uniprot_accession UniProt accessions

action_type Type of drug-target action

mechanism_of_action Mechanism of action descriptions

level5 ATC level 5 codes

atc_name WHO ATC classification names

level1 ATC level 1 codes

level2 ATC level 2 codes

level3 ATC level 3 codes

level4 ATC level 4 codes
warning_type Drug warning types
warning_class Drug warning classes
warning_description Drug warning descriptions
warning_country Country associated with warning
direct_interaction Indicator for direct drug–target interaction

Details

The dataset is denormalized to **one row per ChEMBL drug** (`drug_chembl_id`). Many-to-many relationships (e.g. targets, indications, warnings, ATC codes) are collapsed using semicolon-separated values while preserving first-seen order.

Drug indications

Clinical indications are represented using **four aligned columns**:

- `drug_indication`
- `mesh_id`
- `efo_id`
- `max_phase_for_ind`

Each column contains semicolon-separated values where **positional order is meaningful**. Values at the same position across these columns correspond to the same clinical indication.

Indication records are constructed as **atomic tuples** (`drug_indication | mesh_id | efo_id | max_phase_for_ind`) prior to aggregation to ensure strict positional alignment during collapsing.

Drugs are **never filtered** based on missing indication data. For drugs with **no valid clinical indications**, all indication-related columns are returned as **true NA values** (not character strings).

Missing values and collapsing behavior

During aggregation:

- Missing or empty values are excluded from collapsed fields
- First-seen ordering is preserved
- Fields with no non-missing values are returned as NA
- No empty strings or placeholder values are emitted

Data provenance

The data are extracted from the official ChEMBL SQLite database and processed in `data-raw/` using a reproducible, deterministic pipeline. The raw ChEMBL database file is not distributed with this package.

Source

<https://www.ebi.ac.uk/chembl/>

cmap_method

*CMAP Method for Signature-Based Drug Searching***Description**

Performs a gene expression signature-based drug repurposing method using the CMAP algorithm. It identifies drugs associated with the provided up-regulated and down-regulated gene sets.

Usage

```
cmap_method(object = NULL, upset, downset, ref_db, chunk_size = 5000)
```

Arguments

object	An optional ‘SignatureBased’ object. If provided, it will be updated with the CMAP search results. Defaults to ‘NULL’.
upset	A character vector of up-regulated gene Entrez IDs.
downset	A character vector of down-regulated gene Entrez IDs.
ref_db	A character string specifying the reference database. Valid values are “cmap” or “lincs2”.
chunk_size	The size of chunks for processing large gene sets. Default is ‘5000’.

Details

The function takes up-regulated (‘upset’) and down-regulated (‘downset’) gene IDs, queries a reference database, and performs a CMAP search to identify relevant drugs. The results are returned as a ‘SignatureBasedDrugSearching’ object, or an updated version of it if provided.

Value

A ‘SignatureBased’ object containing:

signature_result	A data frame with the CMAP method results.
query	A list containing the input ‘upset’ and ‘downset’.
signature_method	The method used, “CMAP”.
refdb	The reference database used for the search.

Functions

- `cmap_method()`: Implements the CMAP method for ‘SignatureBased’.

Examples

```
# Example data
upset <- c("gene1", "gene2", "gene3")
downset <- c("gene4", "gene5", "gene6")
ref_db <- "cmap"

# Perform CMAP method search
result <- cmap_method(upset, downset, ref_db)
```

Consensusplot

*Plot Intersection Heatmap for Top K Drug Hits***Description**

Generates a lower-triangular heatmap showing intersections of top K ranked drugs across multiple methods. The number of intersecting drugs is shown for each pair (all drugs and those with valid 'Status'), with formatted intersection text.

Usage

```
Consensusplot(
  object = NULL,
  drug_ranks_input_data,
  TopK = 100,
  file_type = "pdf",
  file_name,
  width = 20,
  height = 15,
  size = 3
)

## S4 method for signature 'PlotObject'
Consensusplot(object, TopK, size)
```

Arguments

object	Optional 'PlotObject'. If 'NULL', one is created from the provided data.
drug_ranks_input_data	A data frame with 'Drug', 'Status', and numeric method columns.
TopK	Number of top ranked drugs to consider (default = 100).
file_type	Output file format: "pdf", "png", "svg", or "jpeg". Default is "pdf".
file_name	Output file name (without extension).
width, height	Plot dimensions. Default: 20x15.
size	Text size for intersection labels. Default: 3.

Value

A 'ggplot' object.

Correlationplot

*Plot Correlation Matrix for Drug Rank Methods***Description**

The ‘Correlationplot’ function computes and visualizes the correlation matrix for drug ranking scores using the specified correlation method (‘pearson’, ‘kendall’, or ‘spearman’). The heatmap displays the full correlation matrix with correlation coefficients and significance stars below the values.

Usage

```
Correlationplot(
  object = NULL,
  drug_ranks_input_data,
  corr = c("pearson", "kendall", "spearman"),
  file_type = "pdf",
  file_name,
  width = 20,
  height = 15,
  size = 3
)

## S4 method for signature 'PlotObject'
Correlationplot(object, corr, size)
```

Arguments

object	An optional object of class ‘PlotObject’. If ‘NULL’, a new ‘PlotObject’ is created.
drug_ranks_input_data	A data frame containing ‘Drug’ and numeric ranking score columns.
corr	The correlation method to compute. One of “pearson”, “kendall”, or “spearman”. Default “pearson”.
file_type	File format for saving the plot (e.g., “pdf”, “png”, “svg”, “jpeg”). Default “pdf”.
file_name	The output filename (without extension).
width	Plot width (default 20).
height	Plot height (default 15).
size	Text size in the plot (default 3).

Value

A ‘ggplot’ object representing the correlation heatmap.

Examples

```
data <- data.frame(
  Drug = c("Drug1", "Drug2", "Drug3"),
  Method1 = c(10, 20, 30),
  Method2 = c(15, 25, 35)
)
Correlationplot(drug_ranks_input_data = data, file_name = "correlation_heatmap")
```

correlation_method	<i>Correlation Method for Signature-Based Drug Searching</i>
--------------------	--

Description

This function performs drug repurposing using a correlation-based approach. The method calculates the correlation between the input gene expression signature ('query') and the reference database, returning a list of potential drugs.

Usage

```
correlation_method(
  object = NULL,
  signature_matrix,
  ref_db,
  method = "spearman"
)

## S4 method for signature 'SignatureBased'
correlation_method(object)
```

Arguments

object	An optional 'SignatureBased' object. If provided, it will be updated with the correlation search results. Defaults to 'NULL'.
ref_db	A character string specifying the reference database. Valid values are "cmap" or "lincs2".
method	A character string specifying the correlation method to use. Default is "spearman". Other options include "pearson", "kendall", etc.
query	A numeric matrix representing the gene expression signature. The matrix should have one column, with gene identifiers stored in the row names.

Details

The function uses a correlation-based approach (e.g., Spearman correlation) to identify drugs that may be associated with the provided gene expression signature. The correlation method can be customized by specifying the method parameter (default is "spearman"). This method uses the 'signatureSearch' package to search the reference database and return potential drug candidates based on the correlation results.

Value

A ‘SignatureBased’ object containing:

signature_result	A data frame with the correlation method results.
query	The input gene expression signature stored as a matrix.
signature_method	The method used, “Correlation”.
refdb	The reference database used for the search.

Functions

- correlation_method(SignatureBased): Implements the Correlation method for Signature-Based Drug Searching.

Examples

```
# Example data
query_matrix <- matrix(c(1.2, -0.5, 2.0), ncol = 1)
rownames(query_matrix) <- c("gene1", "gene2", "gene3")
ref_db <- "cmap"

# Perform Correlation method search
result <- correlation_method(NULL, query_matrix, ref_db, method = "spearman")
```

CRank	<i>CRank Rank Aggregation Method</i>
-------	--------------------------------------

Description

This method applies the CRank algorithm for aggregating rankings from input data. It prepares the input, runs a Python-based CRank analysis, and returns the results as a RankAggregation object.

Usage

```
CRank(
  object = NULL,
  input_data,
  ties_method = "average",
  prior = 0.002,
  num_bin = 800,
  num_iter = 1000,
  reverse = FALSE
)
```

Arguments

object	An object of class 'RankAggregation' (unused in this method but part of the signature).
input_data	A 'data.frame' containing the data to be ranked. The data frame must have at least two numeric columns.
ties_method	A character string specifying how to handle ties in ranking. Options are "average" or "dense" (default is "average").
prior	A numeric value for the prior used in CRank (default is 0.002).
num_bin	An integer specifying the number of bins to use in the algorithm (default is 800).
num_iter	An integer specifying the number of iterations for the algorithm (default is 1000).
reverse	A logical value indicating whether rankings should be reversed (default is FALSE).

Value

An object of class 'RankAggregation' containing the results of the CRank algorithm.

createGraphNetwork	<i>Create Complete Graph Network</i>
--------------------	--------------------------------------

Description

Integrates PPI and drug-target graphs into a complete network.

Usage

```
createGraphNetwork(  
  drug_node_edge,  
  gene_node_edge,  
  drug_node,  
  gene_node,  
  file_name  
)
```

Arguments

drug_node_edge	Drug node-to-edge mappings.
gene_node_edge	Gene node-to-edge mappings.
drug_node	Drug node data.
gene_node	Gene node data.
file_name	Output file name for the graph network.

createNodeToEdge	<i>Create Node-to-Edge Graphs</i>
------------------	-----------------------------------

Description

Creates node-to-edge mappings for PPI and drug-target networks.

Usage

```
createNodeToEdge(PPI_network, Drug_target_network)
```

Arguments

PPI_network	Protein-protein interaction network.
Drug_target_network	Drug-target interaction network.

Value

A list containing PPI and drug graph components.

create_DrugSearchingPipeline_object	<i>create_DrugSearchingPipeline_object</i>
-------------------------------------	--

Description

Creates a new object of class DrugSearchingPipeline

Usage

```
create_DrugSearchingPipeline_object(
  Raw,
  Processed,
  Rankaggregation,
  DrugAnnotation,
  type,
  ...
)
```

Arguments

Raw	List of raw results from drug searching pipeline.
Processed	List of processed results from drug searching pipeline.
Rankaggregation	List of results after rank aggregation.
DrugAnnotation	List of annotated drug information.
...	Additional parameters passed to the new object.

Value

A new instance of DrugSearchingPipeline.

ctg_studies	<i>ClinicalTrials.gov study metadata dataset</i>
-------------	--

Description

A curated, lightweight dataset of publicly registered clinical studies derived from [ClinicalTrials.gov](https://clinicaltrials.gov). The dataset provides core study-level metadata suitable for large-scale analyses, drug–trial mapping, and integration with external biomedical resources.

Usage

ctg_studies

Format

A tibble with one row per clinical study and the following variables:

NCT Number ClinicalTrials.gov study identifier (NCT ID)

Study Title Brief study title

Study Status Overall recruitment or completion status

Conditions Reported study conditions (semicolon-separated)

Interventions Reported interventions (semicolon-separated)

Phases Clinical trial phase(s)

Study Type Type of study (e.g. Interventional, Observational)

Start Date Study start date

Completion Date Study completion date

Details

To reduce package size and improve performance, long free-text fields (e.g. detailed summaries and outcome narratives) have been intentionally excluded. Users requiring full study descriptions should retrieve them directly from ClinicalTrials.gov using the study identifier.

Conditions and interventions

The Conditions and Interventions columns contain semicolon-separated values as reported by study sponsors. These fields are not normalized to controlled vocabularies (e.g. MeSH or EFO) and may include spelling variants or heterogeneous naming conventions.

Dates

Date fields are stored as character strings in the original ClinicalTrials.gov format. Users may convert them to Date objects for downstream analysis if required.

This dataset is generated from the official ClinicalTrials.gov bulk download and processed in data-raw/ using a reproducible pipeline. The raw source files are not distributed with this package.

Source

<https://clinicaltrials.gov>

Description

The `**Degree Centrality**` method calculates the centrality of nodes in a network based on their degree. It is a straightforward measure that counts the number of direct connections a node has to other nodes in the network.

Degree centrality is particularly useful for identifying highly connected nodes (hubs) in a network. In the context of drug discovery, it helps prioritize drugs or genes based on their direct influence within a protein-protein interaction (PPI) or drug-target network.

Usage

```
Degree_centrality(
    object = NULL,
    PPI_network,
    Drug_target_network,
    Disease_genes,
    max_deg,
    result_size,
    target = "drug",
    include_indirect_drugs = TRUE,
    include_non_approved_drugs = TRUE,
    filter_paths = TRUE
)

## S4 method for signature 'NetworkBased'
Degree_centrality(object)
```

Arguments

object An object of class `NetworkCentrality`, containing information about the PPI network, drug-target network, and disease genes.

Details

The function performs the following steps:

1. Validates the input object and ensures all required components are available.
2. Generates graph files for analysis based on the given PPI network, drug-target network, and disease genes.
3. Converts parameters to Python-compatible types using the `reticulate` package.
4. Executes the Python implementation of the Degree Centrality algorithm.
5. Saves the results to a temporary directory and returns the output.

Value

Returns the results from the Python-based Degree Centrality analysis, including:

- Drug scores ranked by their degree centrality in the network.
- Intermediate files stored in a temporary directory.

Functions

- `Degree centrality(NetworkBased)`: Implements the Degree Centrality calculation for the `NetworkCentrality` object.

Examples

```
## Not run:  
# Assuming 'net_centrality' is a valid object of class 'NetworkCentrality'  
results <- Degree_centrality(net_centrality)  
  
## End(Not run)
```

Diffusion*Diffusion-based Drug Search*

Description

A method for running a diffusion-based drug search algorithm.

Usage

```
Diffusion(  
  object = NULL,  
  PPI_network,  
  Drug_target_network,  
  Disease_genes,  
  include_indirect_drugs,  
  include_non_approved_drugs,  
  ties_method,  
  path  
)
```

Arguments

<code>object</code>	A <code>NetworkBased</code> object.
<code>PPI_network</code>	Protein-protein interaction network.
<code>Drug_target_network</code>	Drug target network.
<code>Disease_genes</code>	Genes associated with a disease.
<code>include_indirect_drugs</code>	Whether to include indirect drugs (default is TRUE).
<code>include_non_approved_drugs</code>	Whether to include non-approved drugs (default is TRUE).
<code>ties_method</code>	Method for handling ties in the diffusion algorithm.
<code>path</code>	Path to the data directory.

Value

A modified `NetworkBased` object with the results of the diffusion search.

Donutplot	<i>Plot Donut Chart for Drug Data in AD Studies</i>
-----------	---

Description

The ‘Donutplot’ function generates a donut chart to visualize the distribution of drug data across different groups. The data is grouped by the ‘Drug’ and ‘Highest Status’ columns, allowing the user to visualize how drugs are distributed across these categories. This function creates a "Pie-Donut" chart, with the drug categories as donut slices and the status or study type as pies within each donut.

Usage

```
Donutplot(
  object = NULL,
  data_df,
  file_type = "pdf",
  file_name,
  width = 20,
  height = 15
)

## S4 method for signature 'PlotObject'
Donutplot(object)
```

Arguments

object	A ‘PlotObject’ containing parameters for the plot.
data_df	A data frame containing the data to be visualized. The dataframe should contain columns for ‘Drug’, ‘Highest Status’, and ‘Studies in AD’ or similar groupings.
file_type	The file format to save the plot (e.g., "pdf", "png", "jpeg"). Default is “pdf”.
file_name	The name of the output file (excluding file extension).
width	The width of the output plot in the specified units. Default is 20.
height	The height of the output plot in the specified units. Default is 15.

Value

- A plot object representing the generated donut chart.
- A plot object representing the generated donut chart.

Examples

```
# Example usage with a dataset containing `Drug` and `Highest Status`
data <- data.frame(
  Drug = c("clozapine", "prochlorperazine", "radicicol", "perphenazine", "trifluoperazine", "menadione"),
  Highest_Status = c("Approved", "Approved", "Investigational", "Approved", "Approved", "Approved"),
  Studies_in_AD = c("Pre-clinical", "Unknown", "Unknown", "Unknown", "Unknown", "Unknown")
)
Donutplot(data_df = data, file_name = "donut_plot")
```

Dowdall	<i>Dowdall Rank Aggregation Method</i>
---------	--

Description

Perform rank aggregation using the Dowdall algorithm. The method ranks numeric columns, scales them to the [0, 1] range, computes a global score, and ranks rows based on this score.

Usage

```
Dowdall(object = NULL, input_data, ties_method = "average", reverse = FALSE)
```

Arguments

object	An object of class 'RankAggregation' (optional).
input_data	A 'data.frame' with at least two numeric columns for aggregation.
ties_method	Method to handle ties ("average", "first", "last", "random", "dense"). Default is "average".
reverse	Logical. Reverse the ranking order? Default is 'FALSE'.

Value

An object of class 'RankAggregation' with results.

Dowdall, RankAggregation-method
<i>Dowdall for a Missing Object (Default Creation)</i>

Description

Dowdall for a Missing Object (Default Creation)

Usage

```
## S4 method for signature 'RankAggregation'
Dowdall(object)
```

drugNetworkPipeline *Drug Network Analysis Pipeline*

Description

Performs a comprehensive network-based drug search using centrality, proximity, and diffusion-based methods, then aggregates results using CRank, Dowdall, and RRA algorithms.

Usage

```
drugNetworkPipeline(
  PPI_network,
  Drug_target_network,
  Disease_genes,
  include_indirect_drugs = TRUE,
  include_non_approved_drugs = TRUE,
  hub_penalty = 0.3,
  damping_factor = 0.95,
  result_size = 100,
  nsims = 1000,
  ncpus = 5,
  prior = 0.02,
  num_bin = 200,
  ties_method = "max",
  path
)
```

Arguments

PPI_network	A protein-protein interaction network (graph object or adjacency matrix)
Drug_target_network	A drug-target interaction network (graph object or adjacency matrix)
Disease_genes	Vector of disease-associated gene identifiers
include_indirect_drugs	Logical; include drugs that indirectly target disease genes. Default TRUE
include_non_approved_drugs	Logical; include non-approved drugs. Default TRUE
hub_penalty	Numeric; hub penalty for centrality methods. Default 0.3
damping_factor	Numeric; damping factor for TrustRank. Default 0.95
result_size	Integer; maximum number of results per method. Default 100
nsims	Integer; number of simulations for network proximity. Default 1000
ncpus	Integer; number of CPU cores. Default 5
prior	Numeric; prior for CRank. Default 0.02
num_bin	Integer; number of bins for CRank. Default 200
ties_method	Character; ranking ties method ("max", "min", "dense"). Default "max"
path	Character; output directory for diffusion results.

Value

A list containing:

- ‘DrugSearching’: Centrality, proximity, and diffusion results
- ‘RankAggregation’: Aggregated CRank, Dowdall, and RRA rankings
- ‘Network_Harmonized’: Final harmonized table across all methods

DrugSearchingPipeline-class

DrugSearchingPipeline Class Definition

Description

A class to manage drug signature searching and processing.

drugSignaturePipeline *Drug Signature-based Pipeline*

Description

This function performs a complete drug signature analysis pipeline using differential expression data. It supports multiple (CMAP, LINCS, gCMAP, Correlation) methods, drug synonym filtering, parallel computation, and integrates results via three rank aggregation algorithms.

Usage

```
drugSignaturePipeline(
  signature_input,
  padj = NULL,
  trend = "down",
  Drug_name_synonym = NULL,
  ties_method = "max",
  prior = 0.02,
  num_bin = 200,
  worker = 1
)
```

Arguments

signature_input	Data frame with columns: ‘Entrez’ and ‘FC’.
padj	Numeric adjusted p-value threshold. Default: NULL.
trend	Character filter: "up", "down", or NULL. Default: "down".
Drug_name_synonym	Data frame mapping drugs to synonyms.
ties_method	Method for breaking ties ("max", "min", or "dense"). Default: "max".
prior	Numeric prior for CRank aggregation. Default: 0.02.
num_bin	Number of bins for CRank aggregation. Default: 200.
worker	Number of parallel workers. Default: 1.

Value

A list containing raw, processed, and aggregated drug ranks and drug annotations.

Enrichmentplot

Plot Enrichment Terms from Gene Ontology (GO) Analysis

Description

‘Enrichmentplot’ visualizes GO enrichment results from a provided data frame. Users can choose a dot plot or bar plot to display GeneRatio or Overlap count against GO terms, colored by Adjusted P-value.

Usage

```
Enrichmentplot(
  object = NULL,
  go_terms_df,
  showCategory = 10,
  plottype = c("dotplot", "barplot"),
  file_type = "pdf",
  file_name,
  width = 20,
  height = 15,
  size = 3
)

## S4 method for signature 'PlotObject'
Enrichmentplot(
  object = NULL,
  go_terms_df,
  showCategory = 10,
  plottype = c("dotplot", "barplot"),
  file_type = "pdf",
  file_name,
  width = 20,
  height = 15,
  size = 3
)
```

Arguments

object	Optional ‘PlotObject’. If ‘NULL’, a new one is created from ‘go_terms_df’.
go_terms_df	Data frame with columns ‘Term’, ‘ont’, ‘Adjusted.P.value’, ‘Overlap’ from GO analysis.
showCategory	Number of top categories to display (default: 10).
plottype	Type of plot: “dotplot” or “barplot” (default: “dotplot”).
file_type	Output file format: “pdf”, “png”, “jpeg” (default: “pdf”).
file_name	Output file name (without extension).
width	Plot width (default: 20).
height	Plot height (default: 15).
size	Text label size (default: 3).

Value

A ‘ggplot’ object representing the enrichment plot.

Examples

```
data <- data.frame(
  Term = c("Term1", "Term2", "Term3"),
  ont = c("BP", "MF", "CC"),
  Adjusted.P.value = c(0.01, 0.05, 0.001),
  Overlap = c("5/10", "8/20", "15/30")
)
Enrichmentplot(go_terms_df = data, file_name = "enrichment_plot")
```

gcmmap_method

*gCMAP Method for Signature-Based Drug Searching***Description**

Performs a gene expression signature-based drug repurposing method using the gCMAP algorithm. It identifies drugs associated with the provided gene expression signature.

Usage

```
gcmmap_method(
  object = NULL,
  signature_matrix,
  ref_db,
  higher = NULL,
  lower = NULL,
  padj = NULL
)

## S4 method for signature 'SignatureBased'
gcmmap_method(object)
```

Arguments

object	An optional ‘SignatureBasedDrugSearching’ object. If provided, it will be updated with the gCMAP search results. Defaults to ‘NULL’.
signature_matrix	A numeric matrix with one column, where the row names represent gene identifiers and the matrix values represent the associated gene expression levels.
ref_db	A character string specifying the reference database. Valid values are “cmap” or “lincs2”.
higher	A numeric value representing the threshold for selecting higher-expressed genes. Default is ‘NULL’.
lower	A numeric value representing the threshold for selecting lower-expressed genes. Default is ‘NULL’.
padj	A numeric value representing the adjusted p-value threshold for filtering results. Default is ‘NULL’.

Details

This method uses the gCMAP algorithm to perform drug repurposing based on the provided ‘signature_matrix’. The function can also take additional parameters such as ‘higher’, ‘lower’, and ‘padj’ for fine-tuning the results.

Value

A ‘SignatureBasedDrugSearching’ object containing:

signature_result	A data frame with the gCMAP method results.
query	The input gene expression signature stored as a matrix.
signature_method	The method used, “gCMAP”.
refdb	The reference database used for the search.

Functions

- `gcmmap_method(SignatureBased)`: Implements the gCMAP method for ‘SignatureBased’.

Examples

```
# Example data
signature_matrix <- matrix(c(1.2, -0.5, 2.0), ncol = 1)
rownames(signature_matrix) <- c("gene1", "gene2", "gene3")
ref_db <- "cmap"

# Perform gCMAP method search
result <- gcmmap_method(NULL, signature_matrix, ref_db, higher = 1.5, lower = -1.0, padj = 0.05)
```

generateGraphFiles	<i>Generate Graph Files</i>
--------------------	-----------------------------

Description

Generates the necessary files for running the TrustRank algorithm.

Usage

```
generateGraphFiles(temp_dir, PPI_network, Drug_target_network, Disease_genes)
```

Arguments

temp_dir	Temporary directory for storing intermediate files.
PPI_network	Protein-protein interaction network.
Drug_target_network	Drug-target interaction network.
Disease_genes	Disease-related genes.

Value

A list of file paths for the graph components.

`get_drug_adverse_events`*Extract drug adverse events and toxicity information*

Description

Extracts adverse event, warning, and toxicity information for drugs from supported annotation sources. The function accepts a character vector of drug identifiers or a one-column data frame containing drug identifiers and returns a `DrugAnnotation` object with safety-related information stored in the `result` slot.

Usage

```
get_drug_adverse_events(  
  object = NULL,  
  drugs,  
  source = c("OpenTargets", "ChEMBL")  
)
```

Arguments

<code>object</code>	An optional <code>DrugAnnotation</code> object. If <code>NULL</code> , a new object is created internally using the provided drug input.
<code>drugs</code>	A character vector of drug identifiers or a one-column data frame containing drug identifiers.
<code>source</code>	Character string specifying the annotation source. One of "OpenTargets" or "ChEMBL".

Details

Drug matching is performed in a case-insensitive manner based on drug names.

Supported sources include Open Targets and ChEMBL.

The resulting data frame stored in the `result` slot contains:

Drug Drug identifier provided as input

event Reported adverse event or safety concern

warning_type Regulatory warning category (e.g., Black Box, Withdrawn)

toxicity_class Toxicity classification, when available

description Additional descriptive information, when available

Not all drugs have associated safety annotations. Missing values are retained to reflect source data completeness.

Value

A `DrugAnnotation` object with the `result` slot populated by a data frame containing adverse event and toxicity information for the input drugs.

Examples

```
## Not run:
## Using a character vector
get_drug_adverse_events(
  drugs = c("CETIRIZINE", "PENTAZOCINE"),
  source = "OpenTargets"
)

## Using a one-column data frame
drug_df <- data.frame(drug = c("PHENYLBUTAZONE", "DICLOFENAC SODIUM"))
get_drug_adverse_events(
  drugs = drug_df,
  source = "ChEMBL"
)

## End(Not run)
```

get_drug_atc

*Extract ATC classification for drugs***Description**

Extracts Anatomical Therapeutic Chemical (ATC) classification information for drugs from supported annotation sources. The function accepts a character vector of drug identifiers or a one-column data frame containing drug identifiers and returns a DrugAnnotation object with ATC classification information stored in the result slot.

Usage

```
get_drug_atc(object = NULL, drugs, source = c("ChEMBL", "WHO"))
```

Arguments

object	An optional DrugAnnotation object. If NULL, a new object is created internally using the provided drug input.
drugs	A character vector of drug identifiers or a one-column data frame containing drug identifiers.
source	Character string specifying the annotation source. One of "ChEMBL" or "WHO".

Details

Drug matching is performed in a case-insensitive manner based on drug names.

Supported sources include ChEMBL and WHO ATC classification data.

The resulting data frame stored in the result slot contains ATC codes and corresponding descriptions at multiple hierarchy levels (level 1–5), depending on the annotation source.

Multiple ATC classifications may be associated with a single drug.

Value

A DrugAnnotation object with the result slot populated by a data frame containing ATC classification information for the input drugs.

Examples

```
## Not run:
## Using a character vector
get_drug_atc(
  drugs = c("Imatinib", "Gefitinib"),
  source = "ChEMBL"
)

## Using a one-column data frame
drug_df <- data.frame(drug = c("CETIRIZINE", "PENTAZOCINE"))
get_drug_atc(
  drugs = drug_df,
  source = "WHO"
)

## End(Not run)
```

get_drug_bbb

Extract blood–brain barrier permeability information for drugs

Description

Extracts blood–brain barrier (BBB) permeability classification for drugs using curated BBB annotation data. The function accepts a character vector of drug identifiers or a one-column data frame containing drug identifiers and returns a DrugAnnotation object with BBB classification information stored in the result slot.

Usage

```
get_drug_bbb(object = NULL, drugs)
```

Arguments

object	An optional DrugAnnotation object. If NULL, a new object is created internally using the provided drug input.
drugs	A character vector of drug identifiers or a one-column data frame containing drug identifiers.

Details

Drug matching is performed in a case-insensitive manner based on drug names.

The resulting data frame stored in the result slot contains:

Drug Drug identifier provided as input

BBB_class Blood–brain barrier permeability classification

Multiple BBB classifications may be associated with a single drug and are concatenated using a semicolon.

Value

A DrugAnnotation object with the result slot populated by a data frame containing drug identifiers and their associated blood–brain barrier permeability classification.

Examples

```
## Not run:
## Using a character vector
get_drug_bbb(
  drugs = c("CETIRIZINE", "PENTAZOCINE")
)

## Using a one-column data frame
drug_df <- data.frame(drug = c("CETIRIZINE", "PENTAZOCINE"))
get_drug_bbb(
  drugs = drug_df
)

## End(Not run)
```

get_drug_indications	<i>Extract drug indication information</i>
----------------------	--

Description

Extracts therapeutic indication information for drugs from supported annotation sources. The function accepts a character vector of drug identifiers or a one-column data frame containing drug identifiers and returns a DrugAnnotation object with indication information stored in the result slot.

Usage

```
get_drug_indications(
  object = NULL,
  drugs,
  source = c("OpenTargets", "ChEMBL", "DrugRepurposingHub")
)
```

Arguments

object	An optional DrugAnnotation object. If NULL, a new object is created internally using the provided drug input.
drugs	A character vector of drug identifiers or a one-column data frame containing drug identifiers.
source	Character string specifying the annotation source. One of "OpenTargets", "ChEMBL", or "DrugRepurposingHub".

Details

Drug matching is performed in a case-insensitive manner based on drug names.

Supported sources include Open Targets, ChEMBL, and the Drug Repurposing Hub.

The resulting data frame stored in the `result` slot contains:

Drug Drug identifier provided as input

indication Disease or therapeutic indication

max_phase Maximum clinical development phase associated with the indication

Indication and phase information are harmonized across sources where possible. Multiple indications may be associated with a single drug.

Value

A `DrugAnnotation` object with the `result` slot populated by a data frame containing drug indication information for the input drugs.

Examples

```
## Not run:
## Using a character vector
get_drug_indications(
  drugs = c("Imatinib", "Gefitinib"),
  source = "ChEMBL"
)

## Using a one-column data frame
drug_df <- data.frame(drug = c("CETIRIZINE", "PENTAZOCINE"))
get_drug_indications(
  drugs = drug_df,
  source = "DrugRepurposingHub"
)

## End(Not run)
```

`get_drug_moa`*Extract mechanism of action for drugs*

Description

Extracts mechanism of action (MoA) annotations for drugs from supported annotation sources. The function accepts a character vector of drug identifiers or a one-column data frame containing drug identifiers and returns a `DrugAnnotation` object with MoA information stored in the `result` slot.

Usage

```
get_drug_moa(
  object = NULL,
  drugs,
  source = c("OpenTargets", "ChEMBL", "DrugRepurposingHub")
)
```

Arguments

object	An optional DrugAnnotation object. If NULL, a new object is created internally using the provided drug input.
drugs	A character vector of drug identifiers or a one-column data frame containing drug identifiers.
source	Character string specifying the annotation source. One of "OpenTargets", "ChEMBL", or "DrugRepurposingHub".

Details

Drug matching is performed in a case-insensitive manner using the selected annotation source. The interpretation of drug identifiers depends on the selected source.

Supported sources include Open Targets, ChEMBL, and the Drug Repurposing Hub.

The resulting data frame stored in the `result` slot contains:

Drug Drug identifier provided as input

mechanismOfAction Mechanism of action annotation

Multiple mechanisms of action may be associated with a single drug, depending on the annotation source.

Value

A DrugAnnotation object with the `result` slot populated by a data frame containing drug identifiers and their associated mechanisms of action.

Examples

```
## Not run:
## Using a character vector
get_drug_moa(
  drugs = c("Imatinib", "Gefitinib"),
  source = "ChEMBL"
)

## Using a one-column data frame
drug_df <- data.frame(drug = c("CETIRIZINE", "PENTAZOCINE"))
get_drug_moa(
  drugs = drug_df,
  source = "ChEMBL"
)

## End(Not run)
```

get_drug_status	<i>Retrieve drug regulatory or development status</i>
-----------------	---

Description

Extracts approval or development status information for specified drugs from supported annotation sources. The returned information depends on the selected source.

Usage

```
get_drug_status(object = NULL, drugs, source = c("OpenTargets", "ChEMBL"))
```

Arguments

object	An object of class <code>DrugAnnotation</code> or <code>NULL</code> . If <code>NULL</code> , a new object is created internally.
drugs	A character vector of drug names or a one-column data frame containing drug names.
source	Character string specifying the data source to use. One of "OpenTargets" or "ChEMBL".

Details

This function is implemented as an S4 generic and currently supports objects of class `DrugAnnotation`.

When `source = "OpenTargets"`, the returned fields include regulatory approval status and year of first approval. When `source = "ChEMBL"`, the returned field reflects the maximum clinical development phase recorded in ChEMBL.

Drug matching is performed using case-insensitive string matching on drug names. No synonym expansion or ontology normalization is applied.

Value

A `DrugAnnotation` object whose `result` slot contains drug status information joined to the input drugs.

See Also

[OpenTargets](#), [ChEMBL](#)

get_drug_synonyms	<i>Extract synonyms for drugs</i>
-------------------	-----------------------------------

Description

Extracts alternative names and synonyms for drugs from supported annotation sources. The function accepts a character vector of drug identifiers or a one-column data frame containing drug identifiers and returns a DrugAnnotation object with synonym information stored in the result slot.

Usage

```
get_drug_synonyms(object = NULL, drugs, source = c("OpenTargets", "ChEMBL"))
```

Arguments

object	An optional DrugAnnotation object. If NULL, a new object is created internally using the provided drug input.
drugs	A character vector of drug identifiers or a one-column data frame containing drug identifiers.
source	Character string specifying the annotation source. One of "OpenTargets" or "ChEMBL".

Details

Drug matching is performed in a case-insensitive manner based on drug names.

Supported sources include Open Targets and ChEMBL.

The resulting data frame stored in the result slot contains:

Drug Drug identifier provided as input

synonym Alternative drug name or synonym

Synonym information may include chemical names, brand names, or other alternative identifiers, depending on the annotation source.

Multiple synonyms may be associated with a single drug.

Value

A DrugAnnotation object with the result slot populated by a data frame containing drug synonym information for the input drugs.

Examples

```
## Not run:
## Using a character vector
get_drug_synonyms(
  drugs = c("Imatinib", "Gefitinib"),
  source = "ChEMBL"
)

## Using a one-column data frame
drug_df <- data.frame(drug = c("CETIRIZINE", "PENTAZOCINE"))
```



```

get_drug_synonyms(
  drugs = drug_df,
  source = "OpenTargets"
)

## End(Not run)

```

get_drug_trials

Retrieve clinical trials involving specified drugs

Description

Extracts clinical trial records from the internally bundled ClinicalTrials.gov dataset that involve one or more specified drugs. Drug matching is performed against the Interventions field using pattern matching. Optionally, results may be restricted to trials associated with a given disease or condition.

Usage

```
get_drug_trials(object = NULL, drugs, condition = NULL, ignore_case = TRUE)
```

Arguments

object	An object of class DrugAnnotation or NULL. If NULL, a new object is created internally.
drugs	A character vector of drug names or a one-column data frame containing drug names.
condition	Optional character string specifying a disease or condition to filter trials.
ignore_case	Logical indicating whether matching for drug names and conditions should be case-insensitive. Defaults to TRUE.

Details

This function is implemented as an S4 generic and currently supports objects of class DrugAnnotation.

Value

A DrugAnnotation object whose result slot contains clinical trial records matching the supplied drug(s) and optional condition.

See Also

[ctg_studies](#)

Description

The **Harmonic Centrality** method is used to compute the centrality of nodes in a network, with a focus on drug discovery and repurposing. This method is based on the idea of measuring how central a node is to the network by considering the inverse of the shortest path distances from the node to all other nodes in the graph. It is a variant of closeness centrality that allows for disconnected subgraphs and provides a more robust measure for nodes in complex networks like protein-protein interaction (PPI) networks or drug-target interaction networks.

Harmonic Centrality is particularly useful in identifying drugs or drug targets that are central to a network, even if some nodes are disconnected. It incorporates the structure of both protein-protein and drug-target networks to find drugs that are potentially effective in treating diseases associated with a set of seed genes.

Key Parameters:

- **include_indirect_drugs**: If 'TRUE', drugs that are indirectly connected to seed genes will be included in the search.
- **include_non_approved_drugs**: If 'TRUE', non-approved drugs (such as investigational drugs) will be considered.
- **filter_paths**: If 'TRUE', only certain paths are considered when calculating centrality.
- **hub_penalty**: A penalty factor for hubs (highly connected nodes) in the network. This is useful for avoiding overrepresentation of highly connected genes or drugs.
- **damping_factor**: A factor used in the harmonic centrality calculation to adjust the importance of certain connections.
- **target**: The target type to focus on, such as 'drug' for drug discovery.
- **result_size**: The number of top drugs to return based on their harmonic centrality scores.
- **max_deg**: The maximum degree for nodes in the network. Nodes with a degree greater than this will be excluded from the analysis.

Usage

```
Harmonic_centrality(
  object = NULL,
  PPI_network,
  Drug_target_network,
  Disease_genes,
  hub_penalty,
  max_deg,
  result_size,
  target = "drug",
  include_indirect_drugs = TRUE,
  include_non_approved_drugs = TRUE,
  filter_paths = TRUE
)

## S4 method for signature 'NetworkBased'
Harmonic_centrality(object)
```

Arguments

object	An object of class NetworkCentrality, containing information about the PPI network, drug-target network, and disease genes.
--------	---

Details

The function performs the following steps:

1. Validates the input object and ensures all required components are available.
2. Generates graph files for analysis based on the given PPI network, drug-target network, and disease genes.
3. Converts parameters to Python-compatible types using the `reticulate` package.
4. Executes the Python implementation of the Harmonic Centrality algorithm.
5. Saves the results to a temporary directory and returns the output.

Value

Returns the results from the Python-based Harmonic Centrality analysis, including:

- Drug scores ranked by their harmonic centrality in the network.
- Intermediate files stored in a temporary directory.

Functions

- `Harmonic_centrality(NetworkBased)`: Implements the Harmonic Centrality calculation for the `NetworkCentrality` object.

Examples

```
## Not run:
# Assuming 'net_centrality' is a valid object of class 'NetworkCentrality'
results <- Harmonic_centrality(net_centrality)

## End(Not run)
```

`integrateSignatureNetwork`

Harmonize Signature- and Network-Based Drug Rankings

Description

Integrates the results of signature-based and network-based drug ranking pipelines using CRank, Dowdall, and RRA aggregation algorithms. This function creates a unified harmonized ranking by combining both sources at the rank-aggregation level.

Usage

```
integrateSignatureNetwork(
  signature_res,
  network_res,
  ties_method = "max",
  prior = 0.02,
  num_bin = 200
)
```

Arguments

signature_res	A result object from a signature-based pipeline (e.g., ‘drugSignaturePipeline’) containing ‘RankAggregation’ results for CRank, Dowdall, and RRA methods.
network_res	A result object from a network-based pipeline (e.g., ‘drugNetworkPipeline’) containing ‘RankAggregation’ results for CRank, Dowdall, and RRA methods.
ties_method	Character; method for handling ranking ties (“max”, “min”, “average”, etc.). Default is “max”.
prior	Numeric; prior for CRank. Default is ‘0.02’.
num_bin	Integer; number of bins for CRank discretization. Default is ‘200’.

Value

A nested list containing:

Signature_Based Input signature-based results

Network_Based Input network-based results

Signature_Network Combined aggregation results including:

Signature_Network_CRank CRank aggregation result

Signature_Network_Dowdall Dowdall aggregation result

Signature_Network_RRA RRA aggregation result

Signature_Network_Harmonized Unified merged harmonized table

Examples

```
## Not run:
harmonized <- Harmonize_Signature_Network(signature_res, network_res)

## End(Not run)
```

lincs_method

LINCS Method for Signature-Based Drug Searching

Description

This function performs drug repurposing using the LINCS method to compare gene expression signatures against a reference database (e.g., CMAP). The method uses the LINCS data to identify drugs that might be relevant to the provided gene expression signature.

Usage

```
lincs_method(
  object = NULL,
  upset,
  downset,
  ref_db,
  tau = FALSE,
  sortby = "WTCS",
  chunk_size = 5000,
```

```

    GeneType = "reference"
)

## S4 method for signature 'SignatureBased'
lincs_method(object)

```

Arguments

object	An optional ‘SignatureBasedDrugSearching’ object. If provided, the object is updated with the LINCS search results. Defaults to ‘NULL’.
upset	A character vector containing gene identifiers for the "upset" set of the signature (genes that are upregulated in the signature).
downset	A character vector containing gene identifiers for the "downset" set of the signature (genes that are downregulated in the signature).
ref_db	A character string specifying the reference database for the LINCS search. Valid options include “cmap” or “lincs”.
tau	A logical value (‘TRUE’ or ‘FALSE’) indicating whether to apply the tau ranking during the search. Default is ‘FALSE’.
sortby	A character string specifying the metric to use for sorting the results. Default is “WTCS”. Other options may include alternative sorting metrics depending on the database.
chunk_size	An integer indicating the chunk size for processing the data. Default is ‘5000’.
GeneType	A character string specifying the type of genes used in the reference database search. Default is “reference”.

Details

The function performs a search on the provided gene expression signature (‘upset’, ‘downset’) using the LINCS database, and it calculates similarity scores for the signature against the reference data. Additional options such as sorting by ‘WTCS’ and adjusting the use of tau are available to customize the search. The results are returned as a ‘SignatureBasedDrugSearching’ object.

Value

A ‘SignatureBased’ object that includes:

signature_result	A data frame containing the LINCS search results.
query	A list containing the upset and downset gene sets.
signature_method	A character string specifying the method used for the search, “LINCS”.
refdb	The reference database used for the LINCS search.

Functions

- `lincs_method(SignatureBased)`: Implements the LINCS method for Signature-Based Drug Searching.

Examples

```
# Example data
upset_genes <- c("gene1", "gene2", "gene3")
downset_genes <- c("gene4", "gene5")
ref_db <- "cmap"

# Perform LINCS method search
result <- lincs_method(NULL, upset = upset_genes, downset = downset_genes, ref_db = ref_db, tau = FALSE)
```

listOrMat-class	<i>Signature-Based Drug Searching Class</i>
-----------------	---

Description

A class to store the results of signature-based drug searching using methods like CMAP.

Slots

signature_result A 'data.frame' containing the results of the search.
 query A 'list' containing the up-regulated ('upset') and down-regulated ('downset') gene sets.
 signature_method A 'character' string representing the method used (e.g., "CMAP").
 refdb A 'character' string specifying the reference database used (e.g., "cmap").

NetworkBased-class	<i>Create a SignatureBasedDrugSearching Object</i>
--------------------	--

Description

A helper function to create an instance of the 'SignatureBasedDrugSearching' class.

Arguments

signature_result	A 'data.frame' containing the results of the search.
query	A 'list' containing the up-regulated ('upset') and down-regulated ('downset') gene sets.
signature_method	A 'character' string representing the method used (e.g., "CMAP").
refdb	A 'character' string specifying the reference database used (e.g., "cmap").

Value

A 'SignatureBasedDrugSearching' object.

Description

The Network Proximity method is a network-based drug search algorithm that uses proximity measures in the network to rank drugs. This method explores drug proximity to disease genes using network topological measures and simulations.

Usage

```
Network_proximity(
  object = NULL,
  PPI_network,
  Drug_target_network,
  Disease_genes,
  include_indirect_drugs,
  include_non_approved_drugs,
  nsims,
  ncpus,
  start,
  end,
  random_seed
)
```

Arguments

object An object of class `NetworkBased`, containing the required parameters for running the network proximity method.

PPI_network A data frame representing the protein-protein interaction (PPI) network.

Drug_target_network A data frame representing the drug-target interaction network.

Disease_genes A data frame containing disease-related genes.

include_indirect_drugs Logical, whether to include indirect drugs in the analysis (default is TRUE).

include_non_approved_drugs Logical, whether to include non-approved drugs in the results (default is TRUE).

nsims Integer, the number of simulations to run.

ncpus Integer, the number of CPUs to use for parallel processing.

start Integer, the start index for drugs to consider.

end Integer, the end index for drugs to consider.

random_seed Integer, the random seed for the simulations.

Value

An updated `NetworkBased` object containing the results of the Network Proximity analysis. The results include the ranked list of drugs based on their proximity to the disease genes.

Examples

```
## Not run:
# Create a NetworkBased object and populate it with necessary data.
my_object <- NetworkBased(
  PPI_network = my_ppi_data,
  Drug_target_network = my_drug_target_data,
  Disease_genes = my_disease_genes,
  include_indirect_drugs = TRUE,
  include_non_approved_drugs = FALSE,
  nsims = 1000,
  ncpus = 4,
  start = 1,
  end = 200,
  random_seed = 42
)

# Run the Network Proximity algorithm
result <- Network_proximity(my_object)

## End(Not run)
```

Network_proximity, NetworkBased-method

Network Proximity Method for Network-Based Drug Search

Description

Method for the Network Proximity drug search using a network-based proximity algorithm. The method performs simulations and ranks drugs based on their proximity to disease genes.

Usage

```
## S4 method for signature 'NetworkBased'
Network_proximity(object)
```

Arguments

object	An object of class NetworkBased containing the parameters for the Network Proximity method.
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Value

The same NetworkBased object with the results of the Network Proximity analysis.

OpenTargets*OpenTargets drug annotation dataset*

Description

A harmonised, drug-level dataset derived from the Open Targets Platform. Each row corresponds to a single drug (ChEMBL ID) with integrated information on molecular properties, mechanisms of action, clinical indications, known drug–target evidence, regulatory warnings, and adverse drug reactions.

Usage

```
data(OpenTargets)
```

Format

A data frame with one row per drug and the following columns:

chembl_id ChEMBL drug identifier

name Drug name

drugType Drug modality (e.g. Small molecule, Antibody)

isApproved Logical flag indicating regulatory approval

hasBeenWithdrawn Logical flag indicating withdrawal status

blackBoxWarning Logical flag for black-box warnings

yearOfFirstApproval Year of first regulatory approval

maximumClinicalTrialPhase Maximum clinical trial phase reached

canonicalSmiles Canonical SMILES string

inchiKey InChIKey

mechanismOfAction Semicolon-separated mechanisms of action

actionType Semicolon-separated action types

synonyms Semicolon-separated drug synonyms

efoName Semicolon-separated EFO disease names

disease_id Semicolon-separated disease identifiers

warningTypes Regulatory warning types

toxicityClasses Toxicity classes

warningCountries Countries issuing warnings

adverseEvents Reported adverse events

Details

High-cardinality relationships (e.g. mechanisms, diseases, targets) are stored as semicolon-separated character fields.

Source

Data were obtained from the Open Targets Platform: <https://platform.opentargets.org/>

OptimizeParameters	<i>Optimize Parameters for Network-Based Drug Ranking Methods</i>
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Description

Performs systematic parameter optimization for drug ranking methods such as TrustRank and Harmonic Centrality. Evaluates combinations of hub penalties and damping factors (for TrustRank) and returns results for each parameter set.

Usage

```
OptimizeParameters(
  PPI_network,
  Drug_target_network,
  Disease_genes,
  hub_penalty = seq(0, 1, 0.1),
  damping_factor = seq(0, 1, 0.1),
  cancer_drugs = NULL,
  positive_drugs = NULL,
  negative_drugs = NULL,
  method = c("TrustRank", "Harmonic_centrality"),
  result_size = NULL,
  ncpus = 8
)
```

Arguments

PPI_network	Protein-protein interaction network (graph object or adjacency matrix).
Drug_target_network	Drug-target interaction network (graph object or adjacency matrix).
Disease_genes	Vector of disease-associated gene identifiers.
hub_penalty	Numeric vector of hub penalty values to test. Default: 'seq(0, 1, 0.1)'.
damping_factor	Numeric vector of damping factors (TrustRank only). Default: 'seq(0, 1, 0.1)'.
cancer_drugs	Vector of drugs known for the disease of interest (used for benchmarking).
positive_drugs	Vector of positive-control drugs.
negative_drugs	Vector of negative-control drugs.
method	Character; which ranking method to optimize. One of '"TrustRank"' or '"Harmonic_centrality"'.
result_size	Integer; maximum number of results per run. Default: number of drugs in 'Drug_target_network'.
ncpus	Integer; number of parallel workers. Default: '8'.

Value

A named list of results, where each element corresponds to a parameter combination and contains the ranked drug results.

plotTopKHits

*Visualize Top K Hits for Methods in Drug Ranking Data***Description**

‘plotTopKHits’ generates a plot showing the top K hits for each method in the input ‘drug_ranks_df’ data frame. Supports “Barplot”, “dotplot”, and “lineplot”.

Usage

```
plotTopKHits(
  object = NULL,
  drug_ranks_df,
  plottype = c("Barplot", "dotplot", "lineplot"),
  file_type = "pdf",
  file_name,
  width = 10,
  height = 10,
  units = "in"
)
```

```
## S4 method for signature 'PlotObject'
plotTopKHits(object, plottype)
```

Arguments

object	Optional ‘PlotObject’. If ‘NULL’, one is created from inputs.
drug_ranks_df	Data frame with ‘Drug’, ‘Status’, and numeric ranking columns.
plottype	Type of plot: “Barplot” (default), “dotplot”, “lineplot”.
file_type	Output file format: “pdf”, “png”, “svg”, “jpeg”. Default “pdf”.
file_name	Output file name (without extension).
width, height	Plot dimensions. Default 10 × 10.
units	Units for dimensions: “in”, “cm”, “mm”, or “px”. Default “in”.

Value

A ‘ggplot’ object.

RankAggregation-class *RankAggregation Class*

Description

A class to store results of rank aggregation analysis.

repurposing_hub_drugs *Drug Repurposing Hub compound annotation dataset*

Description

A curated dataset of small-molecule drugs and tool compounds from the ****Drug Repurposing Hub****, developed by the Broad Institute. The dataset contains compound-level annotations including clinical development phase, mechanism of action, molecular targets, disease areas, and therapeutic indications.

Usage

repurposing_hub_drugs

Format

A data frame with one row per compound and the following variables:

pert_iname Compound or perturbagen name

clinical_phase Clinical development phase (e.g. Preclinical, Phase 1, Launched)

moa Mechanism of action

target Primary molecular target(s)

disease_area Broad disease area classification

indication Reported therapeutic indication

Details

The Drug Repurposing Hub is intended to support research in drug discovery, drug repurposing, and chemical biology. The data are provided for research use and are not intended to inform clinical treatment decisions.

This dataset was downloaded from the Broad Institute on March 24, 2020:

https://s3.amazonaws.com/data.clue.io/repurposing/downloads/repurposing_drugs_20200324.txt

Usage restrictions and disclaimer

- The data are provided for non-commercial research use.
- The Broad Institute does not guarantee or warrant the accuracy of the data.
- Reliance on the Drug Repurposing Hub is at the user's own risk.

Citation

Corsello, S. M., Bittker, J. A., Liu, Z., et al. (2017). ***The Drug Repurposing Hub: a next-generation drug library and information resource***. *Nature Medicine*, 23(4), 405–408. doi:10.1038/nm.4306

Source

The Drug Repurposing Hub, Broad Institute. <http://www.broadinstitute.org/repurposing>

RRA	<i>RRA Rank Aggregation Method</i>
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Description

This method performs rank aggregation using the RRA (ROBUST Rank Aggregation) algorithm. It ranks each numeric column, aggregates the rankings across columns, and computes the final RRA p-value and rank.

Usage

```
RRA(  
  object = NULL,  
  input_data,  
  full = FALSE,  
  exact = FALSE,  
  ties_method = "average",  
  reverse = FALSE  
)
```

Arguments

object	An object of class ‘RankAggregation’ (unused in this method but part of the signature).
input_data	A ‘data.frame’ containing the data to be ranked. The data frame must have at least two numeric columns.
full	A logical indicating whether the full output should be returned. Default is ‘FALSE’.
exact	A logical indicating whether exact p-values will be calculated based on rho score . Default is ‘FALSE’.
ties_method	A character string specifying how to handle ties in ranking. Options are "average", "first", "last", "random", and "dense". Default is "average".
reverse	A logical value indicating whether rankings should be reversed. Default is ‘FALSE’.

Value

An object of class ‘RankAggregation’ containing the results of the RRA rank aggregation.

TrustRank	<i>TrustRank: A Network-Based Drug Search Method</i>
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Description

TrustRank is a node centrality algorithm that ranks nodes in a network based on their connectivity to a set of trusted seed nodes (e.g., proteins selected for analysis). The method is a variant of PageRank, initializing seed node centralities with uniform probabilities and setting non-seed nodes' centralities to zero.

The algorithm iterates until convergence, refining node centralities based on the relationships to seed nodes. In undirected graphs, in-neighbors and out-degree are replaced by neighbors and degree, respectively.

References - Gyöngyi, Z., Garcia-Molina, H., & Pedersen, J. O. (2004). Combating Web Spam with TrustRank. **Proceedings of VLDB Endowment**, 576-587. [DOI:10.1016/B978-012088469-8.50052-8](https://doi.org/10.1016/B978-012088469-8.50052-8)

Usage

```
TrustRank(
  object = NULL,
  PPI_network,
  Drug_target_network,
  Disease_genes,
  hub_penalty,
  damping_factor,
  max_deg,
  result_size,
  target = "drug",
  include_indirect_drugs = TRUE,
  include_non_approved_drugs = TRUE,
  filter_paths = TRUE
)
```

Arguments

object	An object of class <code>NetworkCentrality</code> containing the required inputs:
PPI_network	A data frame representing the protein-protein interaction network.
Drug_target_network	A data frame representing the drug-target interaction network.
Disease_genes	A data frame of disease-related genes.
include_indirect_drugs	Logical, whether to include indirect drugs in the analysis.
include_non_approved_drugs	Logical, whether to include non-approved drugs in the results.
filter_paths	Logical, whether to filter specific paths in the graph.
hub_penalty	Numeric, the penalty applied to network hubs.
damping_factor	Numeric, the damping factor for the TrustRank algorithm.
max_deg	Numeric, the maximum degree for nodes in the graph. Defaults to system max if NULL.
result_size	Numeric, the number of results to return.
target	Character, the target context for the analysis.

Value

A data frame containing the ranked list of drugs from the TrustRank analysis.

Examples

```
## Not run:
# Create a NetworkCentrality object and populate it with necessary data.
my_object <- NetworkCentrality(
  PPI_network = my_ppi_data,
  Drug_target_network = my_drug_target_data,
  Disease_genes = my_disease_genes,
  include_indirect_drugs = TRUE,
  include_non_approved_drugs = FALSE,
  filter_paths = TRUE,
  hub_penalty = 0.5,
  damping_factor = 0.85,
  result_size = 100,
  target = "drug"
)

# Run the TrustRank algorithm
result <- TrustRank(my_object)

## End(Not run)
```

TSEA

*Target Set Enrichment Analysis (TSEA)***Description**

Performs Target Set Enrichment Analysis (TSEA) against Gene Ontology (GO) or KEGG databases.

Usage

```
TSEA(
  object = NULL,
  targetList,
  source = "GO",
  ont = c("BP", "MF", "CC", "ALL"),
  Adjusted.P.value = 0.05
)
```

Arguments

<code>object</code>	A ‘DrugAnnotation’ object (optional). If NULL, a new object is created.
<code>targetList</code>	Character vector of gene symbols for enrichment analysis. Required.
<code>source</code>	Database source: "GO" (default) or "KEGG".
<code>ont</code>	Ontology for GO (only when ‘source = "GO"’). Options: - "BP": Biological Process - "MF": Molecular Function - "CC": Cellular Component - "ALL": All three (default)
<code>Adjusted.P.value</code>	Numeric cutoff for adjusted p-value. Default is 0.05.

Value

A ‘DrugAnnotation’ object with enrichment results in ‘result’.

Examples

```
## Not run:
target_genes <- c("TP53", "BRCA1", "EGFR")
TSEA(targetList = target_genes, source = "GO", ont = "BP", Adjusted.P.value = 0.01)

## End(Not run)
```

validateInputs

Validate Inputs

Description

Validates the required inputs for the TrustRank analysis.

Usage

```
validateInputs(object)
```

Arguments

object An object of class NetworkCentrality.

Value

Throws an error if inputs are invalid, otherwise invisible.

WHO

WHO ATC/DDD Drug Classification Dataset

Description

A harmonised drug-level dataset derived from the World Health Organization (WHO) Anatomical Therapeutic Chemical (ATC) and Defined Daily Dose (DDD) classification system.

Usage

```
data(WHO)
```


Format

A data frame with one row per ATC level-5 code and the following columns:

atc_level1 ATC anatomical main group code

atc_level1_name Name of ATC level 1 group

atc_level2 ATC therapeutic subgroup code

atc_level2_name Name of ATC level 2 group

atc_level3 ATC pharmacological subgroup code

atc_level3_name Name of ATC level 3 group

atc_level4 ATC chemical subgroup code

atc_level4_name Name of ATC level 4 group

atc_level5 ATC chemical substance code

atc_level5_name Name of ATC level 5 substance

ddd Defined Daily Dose (DDD)

uom Unit of measurement for DDD

adm_r Route of administration

note Additional WHO notes

Details

The dataset provides full ATC hierarchy (levels 1–5) for each drug-level ATC code, including ATC names, DDD values, administration routes, and notes.

Source

World Health Organization (WHO) ATC/DDD Index

https://www.whocc.no/atc_ddd_index/

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